

Informal Project Report #1

Constant Linear DC Power Supply (PCB Design)

Spring 2026 Semester

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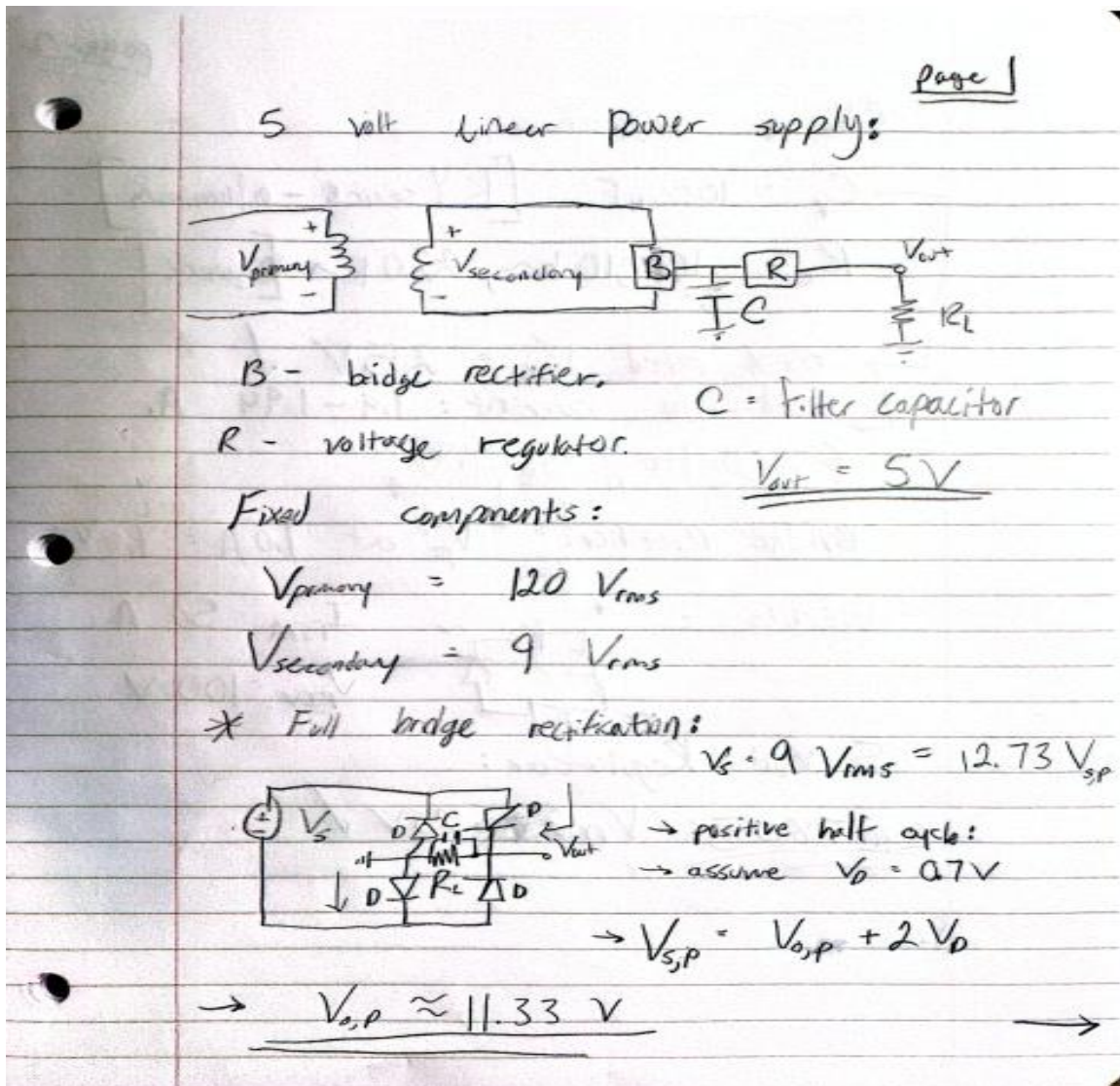
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Introduction:

This informal report details the understanding and analysis of a small project that uses Ki-CAD PCB design software to build a simple constant voltage/DC power supply. Multiple load resistances were used to test the output and response of the power supply(s). The use of full-wave rectification, voltage regulator chips, and filter capacitors are the main design components used alongside a customized PCB board from Ki-CAD. The main objective of this project was to grasp an understanding of Ki-CAD software using a very simple, yet effective constant linear DC power supply with a couple theoretical concepts being verified/analyzed.

Background/Intention:

The main goal of this project was to gain an understanding in basic constant linear power supplies as well as how to design PCB boards using a simple project that can allow use of the basics in Ki-CAD software. Let the basic components be that of a linear DC power supply: AC input, bridge rectifier, filter capacitor, and voltage regulator. Instead of using a Zener diode, an actual voltage regulator chip will be used. Below is idea/note derivation for more detail (Page 1) [1], [2], [3].



Page 1.

Here the assumption for the diode voltage drop is made to be 0.7 V for common silicon diodes [4]. Page 2 verifies the chosen components listed will work with the specifications made (ensuring no excessive loads). Also, the following equations are used to see the relationships between each load resistor chosen (V_d refers to the diode's voltage drop, $V_{m,o}$ refers to the max output voltage, V_r refers to the ripple voltage, R_L refers to the load resistor, C refers to the filter capacitor, and PIV is the max peak inverse voltage of the diode) [4]:

$$PIV = 2V_{m,o} + 2V_d \quad (1)$$

$$V_r = \frac{V_{m,s}}{2fR_L C}, \quad (2)$$

$$i_{D,avg} \approx \frac{1}{\pi} \sqrt{\frac{2V_r}{V_{m,s}}} \frac{V_{m,s}}{R_L} \left(1 + \frac{\pi}{2} \sqrt{\frac{2V_{m,s}}{V_r}} \right), \quad (3)$$

$$i_{D,peak} \approx \frac{V_{m,s}}{R_L} \left(1 + \pi \sqrt{\frac{2V_{m,s}}{V_r}} \right), \quad (4)$$

$$i_{C,peak} \approx \pi \frac{V_{m,s}}{R_L} \sqrt{\frac{2V_{m,s}}{V_r}}, \quad (5)$$

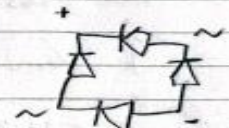
$$i_{C,avg} \approx \frac{\pi}{2} \frac{V_{m,s}}{R_L} \sqrt{\frac{2V_{m,s}}{V_r}}. \quad (6)$$

page 2

Chosen components:

- $C_f = 1000 \mu F$ [KY series - aluminium]
- $R_L = 100 \Omega, 10 k\Omega, 220 k\Omega$ [generic]
- max rated $V_{oc} = 25V$ ✓
- rated ripple current: 1.9 - 1.94 A.
- $F = 60 \text{ Hz}$

Bridge Rectifier: V_F at $1.0 A = 1.0V$

W01G :  $I_{FSM} = 50 A$
 $V_{PIV} = 100 V$

5 Volt Regulator:

LM7805: $V_{max} = 25V$ ✓

MATLAB code below yields the following results (Table 1) with load resistances of 100 Ω , 10 k Ω , and 220 k Ω .

Table 1: Theoretical Data (Verification to Operate)

<i>Parameter</i>	$R_L = 100 \Omega$	$R_L = 10 \text{ k}\Omega$	$R_L = 220 \text{ k}\Omega$
$i_{C,avg}$ (A)	0.8719	0.0872	0.0186
$i_{C,peak}$ (A)	1.7438	0.1744	0.0372
$i_{D,avg}$	0.1280	0.011	0.0001
$i_{D,peak}$ (A)	1.8571	0.1755	0.0372
V_r (V)	0.9442	0.0094	0.0004
PIV (V)	12.03	12.03	12.03

Based on Page 2 above, the components were verified to work with the theoretical data above. Next, an LTspice simulation was built (Figure 1) with the components W01G (bridge rectifier), LM7805 (voltage regulator), and secondary voltage source based off the fixed values. LTspice does not offer a LM7805 or W01G bridge rectifier, however, online Git Hub sources (provided in References) offer an accurate method of implementing simulation [5], [6].

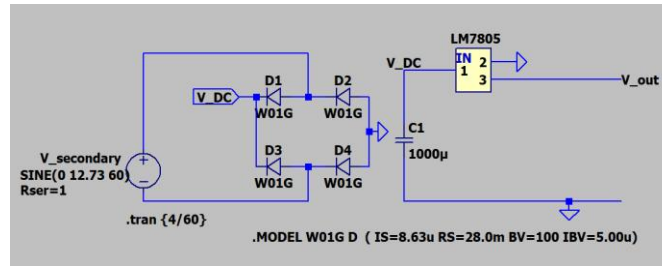


Fig. 1. LTspice simulation.

The simulation appears to work as intended with the following table (Table 2) sharing the simulation values in comparison to the theoretical values. The low resistance ($R_L = 100 \Omega$) error can be described by the theoretical linear approximation. These tables are just to verify operation and are not the main objective of this project.

Table 2: Simulated Data

Parameter	$R_L = 100 \Omega$	$R_L = 10 \text{ k}\Omega$	$R_L = 220 \text{ k}\Omega$
$i_{C,avg}$ (A)	0.0023	0.0	0.0
$i_{C,peak}$ (A)	0.668	0.059	0.048
$i_{D,avg}$	0.0291	0.0085	0.0
$i_{D,peak}$ (A)	0.429	0.062	0.056
V_r (V)	0.261	0.036	0.007
PIV (V)	13.06	13.06	13.06

Table 2 above verifies operation region in simulation where theoretical computation fails. No thresholds or maximum component limitations will cause failure of the project so the Ki-CAD design can commence. Beforehand, figures below (Fig. 2, 3, 4, and 5) share the voltage output of the simulation and current values for each load resistance used above.

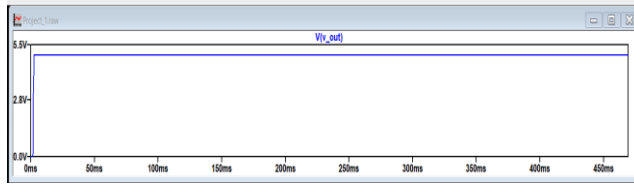


Fig. 2. Verifies DC voltage at 5.0 V.

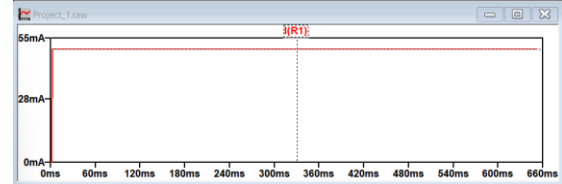


Fig. 3. Verifies DC current at 50.0 mA ($R_L = 100 \Omega$).

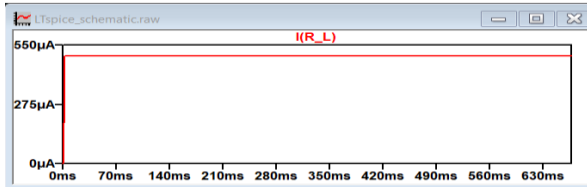


Fig. 4. Verifies DC current at 500.0 μA ($R_L = 10 \text{ k}\Omega$).

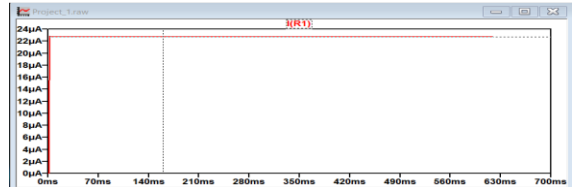


Fig. 5. Verifies DC current at 22.75 μA ($R_L = 220 \text{ k}\Omega$).

Ki-CAD Design:

Using Ki-CAD software and built-in hardware components, the Ki-CAD learning process was straightforward and clear. Ki-CAD software had built-in components that were used in this project; therefore, the schematic was easy to assemble. Screw terminals were used to mimic input and output rails. Figure 6 below shares the finalized schematic. Note that C_2 is a stabilizing capacitor for the output voltage.

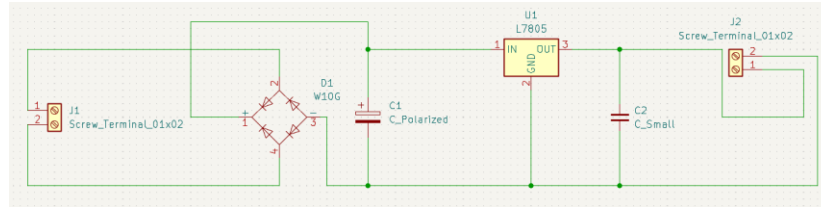


Fig. 6. Ki-CAD schematic.

Ki-CAD internal electrical component checker was used to verify all connections were safe and no obvious short circuits. Next, the PCB editor was utilized to build a custom PCB board that would house every component. Figures 7 and 8 show the finalized layout for this part as well as the 3-D viewer of Ki-CAD software.

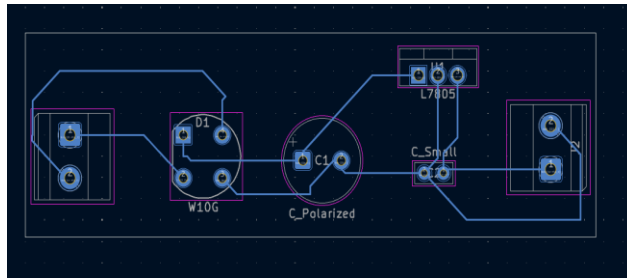


Fig. 6. PCB Layout in Ki-CAD.

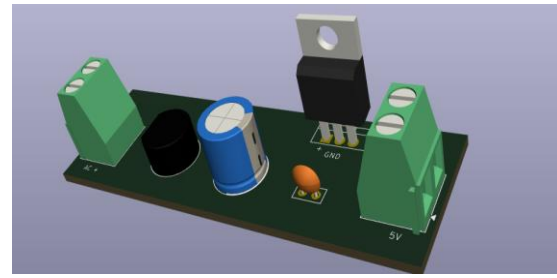


Fig. 7. PCB 3-D Viewer in Ki-CAD.

Only the back of the PCB was utilized in wiring each component. The soldering of the components was done separately for practice in hand-soldering PCBs. The following table (Table 3) summarizes an overall cost of the entire project alongside the exact components used.

Table 3: Materials and Equipment Used

<i>Component</i>	<i>Manufacturer (if applicable)</i>	<i>Cost (\$)</i>	<i>Purchased at</i>
Screw Terminal Block Kit (85 Pcs)	N/A	9.99	Amazon
AC to AC Wall Adapter Transformer (120/240 V to 9 V)	HQRP	22.36	Amazon
PCB Board	JLCPCB	82.93	JLCPCB
22 Gauge PVC 1007 Sold Electric Wire (Red and Black)	BNTECHGO	14.98	Amazon
10 Values Linear Voltage Regulators (50 Pcs)	EEEEEE	13.99	Amazon

Monolithic Ceramic Capacitors (120 Pcs, Value: 0.1 μ F)	Mxutuek	6.99	Amazon
1000 μ F Aluminum Electrolytic Capacitors (Rated 25 V _{DC})	Chemi-Con	1.28	Mouser
W01G Bridge Rectifier	Vishay Semiconductors	1.59	Mouser
Solderless Breadboard	—	8.99	School Lab Kit
Various Resistors (220 k Ω , 100 k Ω , 10 k Ω , 1 k Ω 220 Ω , 100 Ω , 1 Ω)	—	$\approx$ 8.00	School Lab Kit
Light Emitting Diode (Any Color)	—	$\approx$ 2.00	School Lab Kit
Digital Multimeter	—	—	—
Oscilloscope (Optional)	—	—	—

*Datasheets provided in References section (not applicable for 0.1 μ F Cap). Other brands of materials can be used for this experiment.

*Total cost estimated $\approx$ \$200.00 (equipment not factored into cost).

The Ki-CAD model was successfully ordered and soldered with components from Table 4. An LED was just used as a load alongside the resistors for fun. The following pictures (Fig. 8 and 9) share the printed board with the components soldered successfully.



Fig. 8. Rear of PCB Board.

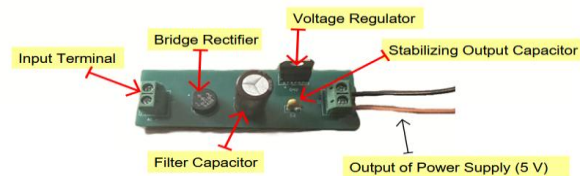


Fig. 9. Front of PCB Board.

Analysis and Conclusion:

The following table (Table 4) displays some of the results from the lab to verify that the power supply worked the same according to the various loads attached at the open circuit output. The following equations were used to compute different power outputs and compare them to ideal computations (using Ohm's Law (8) and ideal values; refer to Page 3 and 4) [4]:

$$P = VI, \quad (7)$$

$$V = IR.$$

(8)

Table 4: Results

Parameter (R_L, Ω)	Computed Power (W)	Measured Power (W)	Theoretical I_{LOAD} (A)	Measured I_{LOAD} (A)
1	25	9.37×10^{-3}	5	93.7×10^{-3}
100	250×10^{-3}	250.5×10^{-3}	50×10^{-3}	50.1×10^{-3}
1,000	25×10^{-3}	25.03×10^{-3}	5×10^{-3}	5×10^{-3}
10,000	2500×10^{-6}	2501×10^{-6}	500×10^{-6}	500.2×10^{-6}
220,000	113.65×10^{-6}	114.5×10^{-6}	22.73×10^{-6}	22.9×10^{-6}
231,101	108.18×10^{-6}	109×10^{-6}	21.64×10^{-6}	21.8×10^{-6}

Data Points to Ensure
Working as Intended:

measured: * open circuit voltage: 4.99 V

R_L :	I_L :	V_L :	P_L :
1 Ω	93.7 mA	0.10 V	9.37 mW
100 Ω	50.1 mA	5.00 V	250.5 mW
1 k Ω	5.0 mA	5.01 V	25.05 mW
10 k Ω	500.2 μ A	5.00 V	2501.0 μ W
220 k Ω	22.9 μ A	5.00 V	114.5 μ W
231,101 Ω	21.8 μ A	5.00 V	109.0 μ W

LED Data:

$V_{LED} = 1.5 V$ at 15 mA. $V_s = 5 V$

$\Rightarrow \frac{V_s - 1.5}{R_{LED}} = 15 \times 10^{-3}$

$R_s = 222 \Omega \approx 220 \Omega$

$\Rightarrow I_L$ at 220 Ω : 9.47 mA (w LED)

$\Rightarrow I_L$ at 220 Ω : 21.82 mA (no LED)

$\Rightarrow V_{LED}$ measured: 2.81 V

$\Rightarrow V_{LED}$ computed: 3.5 V

$\Rightarrow$ LED is on and bright ✓

Ideal:

R_L	I_L	V_L	P_L
1 Ω	5 A	5 V	25 W
100 Ω	50 mA	5 V	250 mW
1 k Ω	5 mA	5 V	1 mW
10 k Ω	500 μ A	5 V	2500 μ W
220 k Ω	22.73 μ A	5 V	113.65 μ W
23,101 Ω	21.64 μ A	5 V	108.18 μ W

Page 4.

Notice that the only discrepancy in the analytical and measured data is the 1 Ω load; the other load resistors appeared to display characteristics that replicate the theoretical data. This confirms that the Ki-CAD PCB design worked successfully in producing a basic constant linear DC power supply. The 1 Ω load resistance appears as a short circuit to the voltage regulator. This causes the regulator to dissipate much of the current output in forms of heat, resulting in no output voltage and corresponding load current. This is shown in Page 3, where the output voltage is essentially 0 V and the output current stabilizes around 93.7 mA. The voltage regulator datasheet (under the 7805) states the max output short circuit current is roughly 750 mA [3]. This is true in this case, where the current begins at approximately 500 mA and steadily decreases until it stabilized to 93.7 mA while continuously dissipating excessive power in forms of heat (thermal throttle) [4]. The voltage regulator got noticeably *hot* during the short circuit test. However, each load resistance onward presented valid results as the output voltage was consistent with a 5.0 V DC output. The current values (using (8)) lined up as well with minimal error. The small linear DC power supply is a success for the objective being to understand the inner working and application of Ki-CAD software and its tools. This project will be a building block in transferring to Altium Designer, a more rigorous application in real-world solutions.

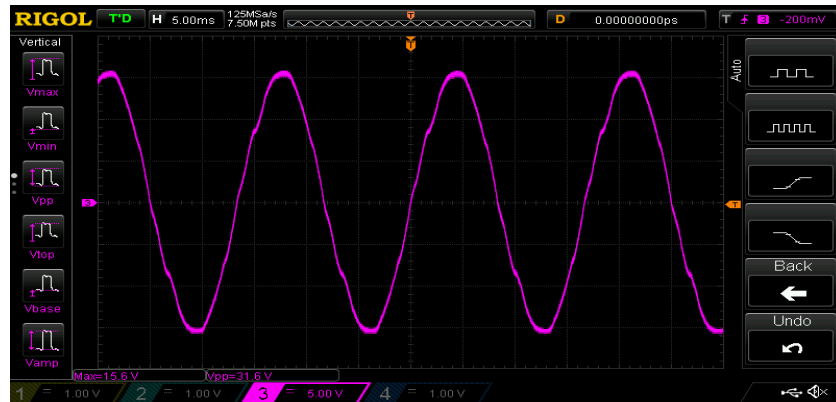
References:

- [1] Nippon Chemi-Con, “KY Series: Aluminum Electrolytic Capacitors,” datasheet, Tokyo, Japan, n.d.
- [2] Vishay Semiconductors, “Glass Passivated Single-Phase Bridge Rectifier,” W005G through W10G Series, Datasheet, Malvern, PA, Jul. 9, 2009.
- [3] Texas Instruments, “ μ A7800 Series: Positive-Voltage Regulators,” SLVS056J, Datasheet, Dallas, TX, May 1976 [Revised May 2003].
- [4] C. Johnson. ECE 3050. Class Lecture, Topic: “Diode/Half Wave Rectification.” Department of Electrical and Computer Engineering, Tennessee Technological University, Cookeville, TN, Mar. 10, 2026.
- [5] Peteut, “diodes_bridge-rectifiers.text,” *spice-models*, Apr. 18, 2025. [Online].

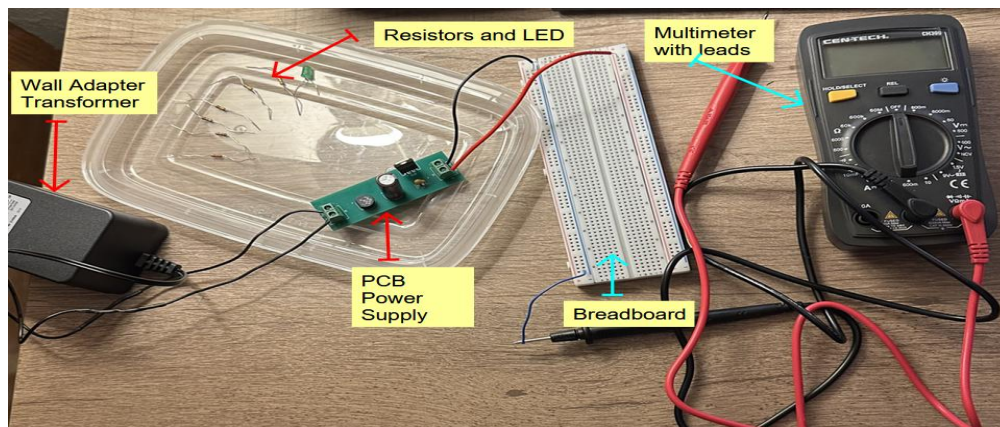
Available: https://github.com/peteut/spice-models/blob/master/diodes/diodes/diodes_bridge-rectifiers.txt. [Accessed May 5, 2026].
- [6] Kafana, “regulators.lib,” *models*, Mar. 17, 2016. [Online].

Available: <https://github.com/kafana/ltspice-misc/blob/master/models/regulators.lib>. [Accessed May 5, 2026].

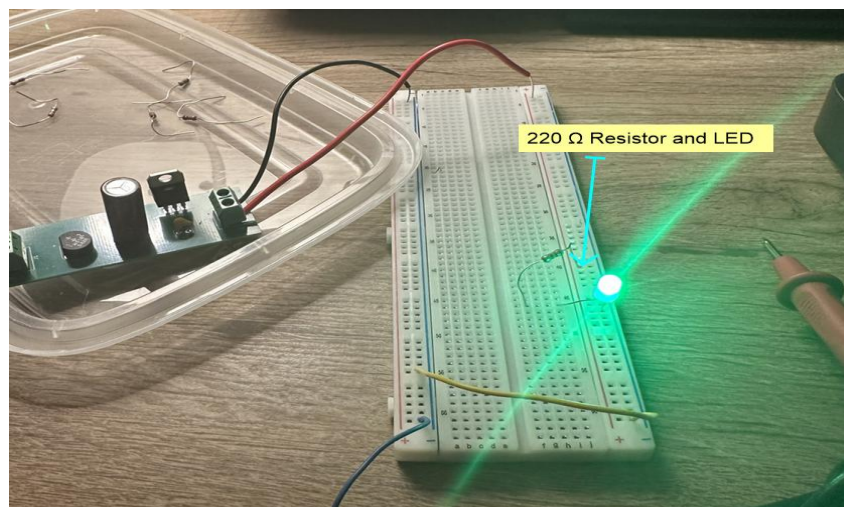
Additional Data (Photos during Analysis):



Oscilloscope waveform of the input AC voltage from the transformer ($V_{s,p} = 15.6\text{ V}$).



Testing Layout of Project.



Last Part (Page 4) in Action.